

ADOPTION OF PREFABRICATED CONSTRUCTION IN BANGLADESH: PROSPECTS AND OBSTACLES

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ABSTRACT

Nowadays, modern construction technology and methods are being utilized throughout the world. In this case, prefabricated construction is one of the efficient methods that are used by both the developed and developing countries. Bangladesh is not an exception to resort this system. The intent of this study is to ascertain and explore the prospects and obstacles of prefabricated construction in Bangladesh. An outspread literature investigation was done to execute a questionnaire survey for the study. Relative Importance Index (RII) was used to analyze and rank the sixteen possible prospects and fifteen obstacles of such construction to be implemented in this country. It is obvious from the analysis that savings in construction time by fast installations, remission of project cost, increase in construction quality, more safety during construction and low construction waste and noise are the main prospects to experience. Besides, unfamiliarity with prefabrication concept, insufficient knowledge on design and execution, popularity of conventional construction, insufficient skilled personnel and need of specialized equipment for installation are the top obstacles to be seen. Finally, this work tends to recommend useful suggestions to lessen the barriers, so that sustainable construction can be experienced along with the amenities of prefabricated construction.

Keywords: Prefabrication; Construction; Prospects; Obstacles; Bangladesh.

INTRODUCTION

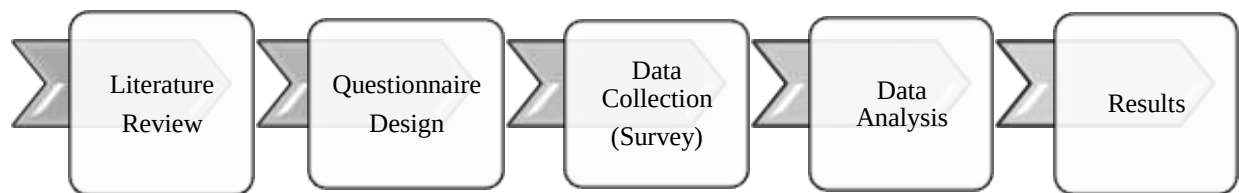
Conventional building construction offers numerous problems like poor safety of workers, excess cost, need of more manpower, long construction period, excess construction waste and so on. That's why modern construction methods and techniques are emerging in worldwide. Being a densely populated country, Bangladesh is dealing with these problems that are experienced regularly. In Bangladesh, the concept of prefabricated construction is comparatively new. Prefabrication generally indicates the process of creating the components of a building inside a factory according to the specified design, then these components are transported on site and gets connected to form a building. Prefabricated construction is somewhat first degree of industrialization, followed by mechanical advancement, automation, robotics and regeneration (Richard, 2005). Also, prefabrication grants a tangible advantage to increase projects' sustainable performance and behavior (Chen et al., 2010). The construction industry of Bangladesh is trying to adopt new construction techniques towards sustainable and effective construction. In this case, prefabrication may become one of the smartest solutions that can combine all possible advantages for the betterment of construction practices.

A mighty excuse for the incuriosity among higher authority and stakeholders to adopt this system is that they face difficulty to find out and finalize the amenities which may add to a project (C. L. Pasquire & Connolly, 2002). But practically, prefabricated construction is such a method that are cost and time effective, provides more durability for long run and minimizes on site construction waste. However,

curiosity to use prefabrication based on publicity and personal choice is common around the world (Idrus, 2002). This system is also applicable in Bangladesh. A true fact is that, the concept and knowledge regarding this method is not popular here. In addition, some problems prevails at the initial phase of such construction, which may look as threatening factors to implement this system. Besides, inadequate importance are applied to the conversations if it's possible to prefabricate a building or parts of a building (C. Pasquire et al., 2005). So, this research focuses on identifying and presenting the most possible prospects and obstacles towards the adoption of prefabrication by the construction industry of Bangladesh. This work may accompany higher authorities, stakeholders, clients and sponsors to understand every factor that are interrelated to prefabricated construction. Finally, this paper recommends some ways to reduce the obstacles so that effective and sustainable construction can be experienced by the adoption of prefabricated construction in Bangladesh.

METHODOLOGY

The necessary data was gathered from research papers and practical inspections. The data was audited to identify the most possible prospects and obstructive factors that are related to prefabricated construction. A good percentage of significant data and information was accumulated particularly from a compatible inquiry and the remaining was stated using available sources of news. Figure 1 shows the entire workflow for this research:



[Fig. 1]: Workflow Diagram

Data Collection

The type and character of this review can be termed as quantitative data collection. In this approach, observation through journals and conference papers, contents seemed to be useful for the design of questionnaire survey. From the effective literature scan, sixteen possible prospects and sixteen crucial obstacles were found towards the adoption of prefabricated construction. Then, the questionnaire was distributed through google form to the learned persons for the completion of the online survey. In this process, the active respondents were architects, engineers, project managers, educators and students from the relevant field. Following tables represent the sample of response collection, where level of each factor was judged based on relative value (1-strongly disagree, 2- disagree, 3-neutral, 4-agree, 5-strongly agree):

Table 1. Prospects of Prefabricated Construction

Prospects	Relative Value				
	1	2	3	4	5
Savings in construction time				*	
More safety during construction			*		
Remission of project cost				*	

Reduction of water & energy consumption					*
Independent of hostile weather			*		
Less construction difficulty		*			
Easy supervision of project			*		
Flexibility in demolition work				*	
Less construction noise and waste			*		
Reuse of components			*		
Increase in construction quality					*
Requirement of less manpower				*	
More fire resistant			*		
Utilization of aesthetic perspectives			*		
Enhanced environmental performance				*	

Table 2. Obstacles of Prefabricated Construction

Obstacles	Relative Value				
	1	2	3	4	5
Popularity of conventional construction				*	
Lack of market awareness					*
Inadequate research and fund				*	
Client's incuriosity with proposal				*	
Problem of leakage			*		
Loss of job scopes			*		
Need of specialized equipment for installation				*	
Necessity of extra care during transportation		*			
Insufficient knowledge on prefab. design and execution			*		
Insufficient skilled personnel					*
High capital cost				*	
High transportation cost				*	
Difficulty in sudden change of design				*	
Inadequate site space			*		
Unfamiliarity with prefab. concept					*

Data Analysis

To expose the prospects and obstacles towards the adoption of prefabricated construction, Relative Importance Index (RII) was used. This helped to identify and rank the factors based on respondents' opinion. Equation (1) displays the mathematical formula for the calculation of Relative Importance Index:

$$\text{Relative Importance Index} = (\sum W) / (AN) \quad (1)$$

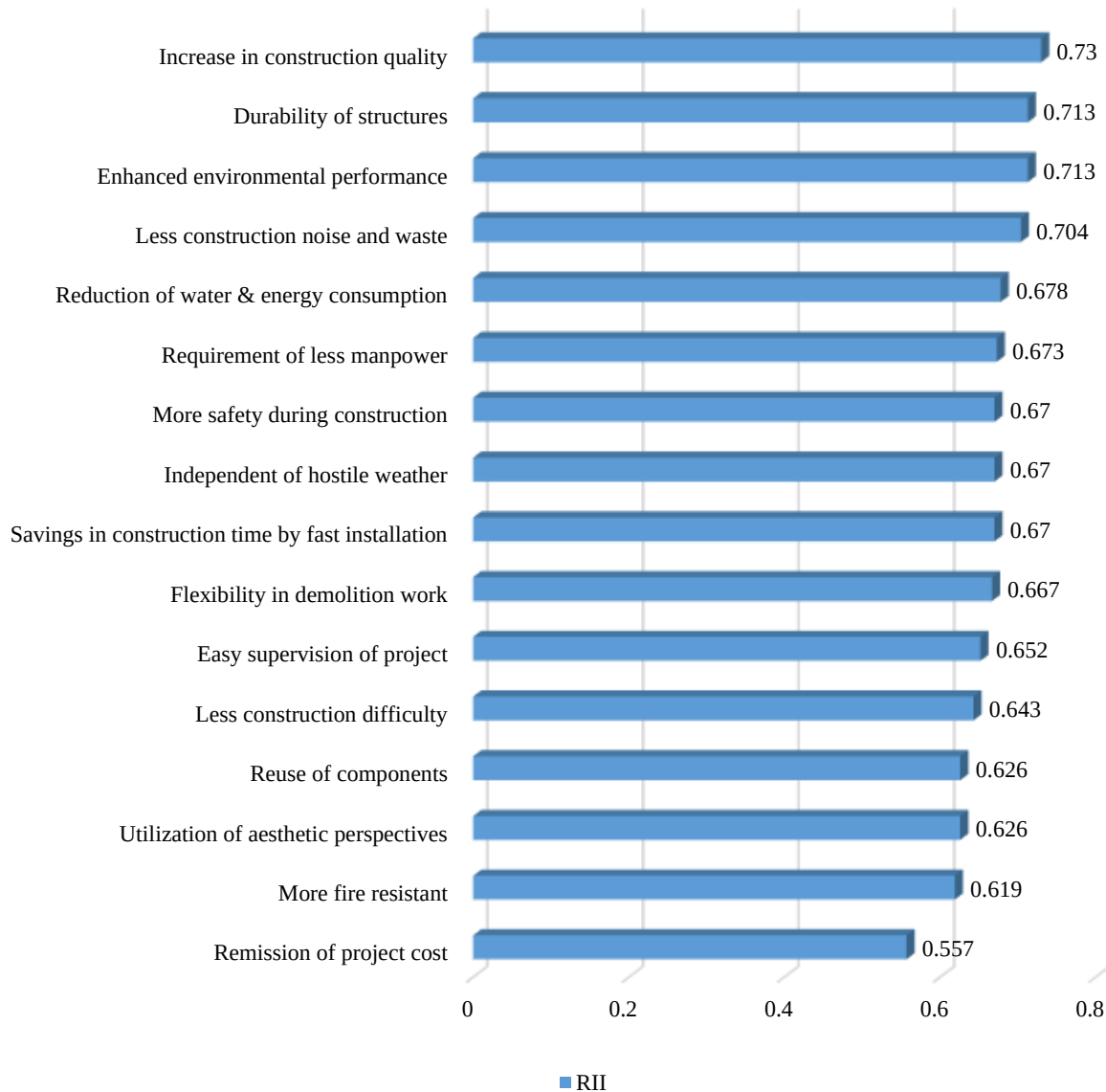
Here, W = Weight given to each factor (Relative value 1 to 5, where 1 indicates the lowest value and 5 indicates the highest).

A = Highest value of range (In this study, A = 5) and,

N = Number of total respondents.

RESULTS AND DISCUSSIONS

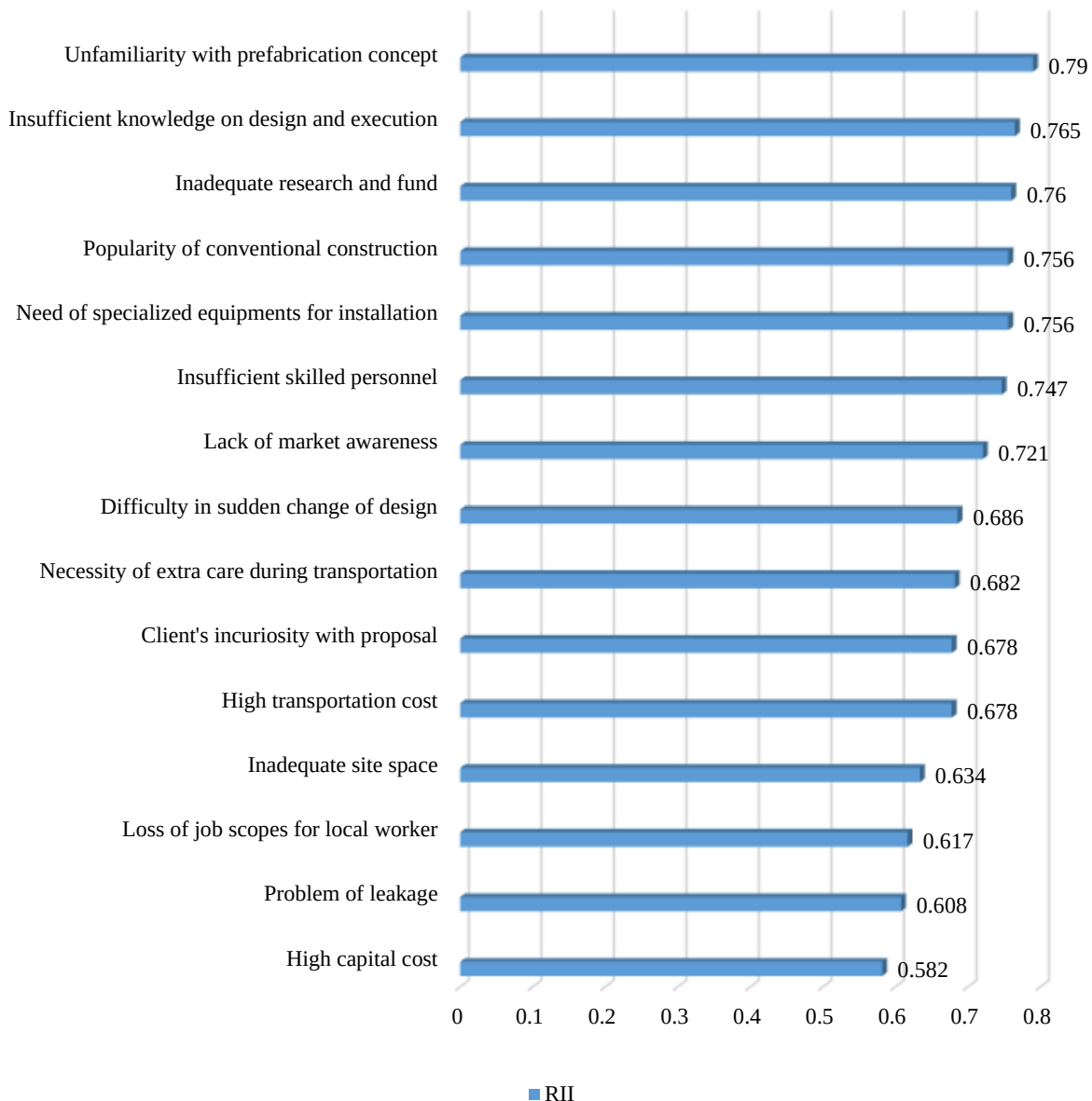
In this study, sixteen major prospects were found towards the adoption of prefabrication in the construction practices of Bangladesh. In terms of Relative Importance Index (RII), these are presented and ranked below:



[Fig. 2]: Prospects of Prefabricated Construction

Above figure disclosed increase in construction quality, durability of structures, enhanced environmental performance, less construction noise and waste, reduction of water and energy consumption and requirement of less manpower as the major prospects of prefabricated construction with respect to Bangladesh. Except these, other factors may contribute significantly. It is obvious from the analysis if prefabricated construction practices can be developed, the country will experience numerous positive changes towards sustainable construction.

Besides, the study sorted fifteen obstacles to adopt this construction system. Figure 3 reveals the degree of obstacles based on the Relative Importance Index (RII).



[Fig. 3]: Obstacles of Prefabricated Construction

Figure 3 displayed unfamiliarity with prefabrication concept, insufficient knowledge on prefabricated design and execution, inadequate research and fund, popularity of conventional construction, need of specialized equipment for installation of components and insufficient skilled personnel as the top obstructions to adopt the prefabricated construction system. To summarize, all of the fifteen obstacles are accountable to resist the adoption of prefabricated construction practice, for which the country may suffer in near future.

CONCLUSIONS

Prefabricated construction practice shaping is a great concern for developing country like Bangladesh, as the country is facing too much environmental contamination for the uncontrolled construction work. That's why the work was undertaken to outline the possible prospects and obstacles towards the adoption of this construction system. On the basis of RII, increase in construction quality, durability of structures, enhanced environmental performance, less construction noise and waste, reduction of water and energy consumption and requirement of less manpower were exhibited as major emerging prospects. In addition, unfamiliarity with prefabrication concept, insufficient knowledge on design and execution, inadequate research and fund, popularity of conventional construction, need of specialized equipment for installation and insufficient skilled personnel were the leading obstacles based on the analysis. To minimize the obstructions, awareness of prefabrication concept and process, long run cost-effectiveness, essential research and funds, availability of equipment and tools and development of engineer, project manager and worker's competence in prefabricated design, construction and execution must be assured. In this manner, the most viable prospects can be used for advanced research and practically Bangladesh may face less difficulty and experience sustainable construction by conquering enormous challenges and obstructions.

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